

SAKSHEE UJJWAL KUMAT

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EDUCATION

Vellore Institute of Technology, Chennai

B.Tech in Computer Science and Engineering Core

2023 – Present

CGPA: 9.05

Srimathi Sundaravalli Memorial School, Chennai

10th Board [CBSE] – 98.4% 12th Board [CBSE] – 95.6%

PROJECTS

SECUREBANK: CONTEXT-AWARE PASSWORDLESS AUTHENTICATION

December 2025 – Present

React, TypeScript, Flask, Python, SQLite, Web Crypto API

- Engineered a **passwordless authentication system** for a banking workflow using **React** and **Flask**, implementing **RSA-2048** device-bound authentication via the **Web Crypto API** with **context-aware nonce binding (SHA-256)** and **TOTP** step-up.
- Improved cryptographic performance, achieving **500 ms authentication latency**, **201 ms RSA key generation**, and **128 ms PBKDF2 derivation**, ensuring secure and responsive login flows.
- Developed a **real-time risk scoring engine (0–100)** using **7 contextual signals** to classify requests into **ALLOW / STEP-UP / DENY** with **1.2 ms computation time**.

BLOCKCHAIN BASED WOMEN SECURITY APP

July 2025 – Present

Flutter, Node.js, Express.js, Solidity, Hardhat, Firebase

- Built a **role-based, decentralized SOS system** with **Flutter** and **Node.js**, enabling incident reporting with **real-time GPS data** for women and backed by **volunteers**.
- Designed and deployed **Solidity smart contracts** on a local **Hardhat Ethereum network** for **immutable, tamper-proof storage**.
- Integrated **Firebase authentication** and backend validation to execute **secure on-chain transactions** via ABI calls.

NEURODECEPT: ML-DRIVEN HONEYPOT OPTIMIZATION

July 2025 – October 2025

Python, Scikit-learn

- Modeled network behavior using **unsupervised anomaly detection (Autoencoders, Isolation Forest, LOF)** and analyzed anomaly score distributions.
- Devised a **risk-based controller** to optimize **honeypot placement** under limited deployment capacity.
- Quantified impact using simulated attack scenarios, achieving **75% diversion** (54/72 attacks) through **data-driven topology adaptation**.

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, Bash

Core: Agile, DSA, OS, Computer Architecture, DBMS, Cloud Architecture Design

Web & Backend Development: JavaScript, TypeScript, React, Flask, FastAPI, Node.js, Express.js, Flutter

Databases: MySQL, SQLite, Firebase

Blockchain & Security: Solidity, Web Crypto API

Cloud & DevOps: IBM Cloud, Git, GitHub Actions, Postman, Docker

LEADERSHIP

Outreach Lead at VITeach

2024 – Present

- Led 4 outreach programs across government schools and collaborated with NGOs to support education for underprivileged students.
- Coordinated volunteering drives and academic support camps, including managing collaborations and event executions.

CERTIFICATIONS & ACHIEVEMENTS

- Generative AI using IBM Watsonx** – Issued by IBM, June 2025
- Aryabhata National Maths Competition by AICTSD (2024)** – Appeared for the final interview round